LAM – Complete Product and Engineering Requirements

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LAM (Learning Adaptation Model) is an independent system that makes an AI tutor progressively better at helping each student learn, remember and apply concepts, without depending on the tutor for every answer.

1. What LAM must ultimately accomplish

The core loop is: **Observe, Diagnose, Teach, Test, Update Learner Model, Select Next Best Action, Revisit Later.**

The LAM PRD already establishes the service boundary, per-concept mastery, diagnosis, adaptive decisions, retrieval scheduling, API integration and a governed shared-model pipeline. This document expands those foundations into a complete product and engineering checklist. Items described as additional requirements are recommendations for completing the specification, not claims that they are already implemented.

LAM must answer seven questions reliably for every learning interaction

1. What does this student appear to understand?
2. What evidence supports that estimate, and how uncertain is it?
3. If the student is struggling, what are the plausible reasons?
4. What should happen next to improve learning?
5. How much assistance should the student receive?
6. Did the intervention improve independent understanding, retention or transfer?
7. When should the system revisit the concept?

A generic chatbot can answer questions. LAM must manage the learning process across interactions and over time.

Core principles

- A correct answer does not prove mastery.
- A wrong answer does not prove a misconception.
- A strategy that works for one student will not necessarily work for another.

2. The complete LAM capability map

These 15 capability areas are the master checklist for the product.

#	Capability	Purpose
01	Evidence collection and event ledger	Capture reliable, structured learning events
02	Learner profile and state engine	Maintain persistent, per-student, per-concept learning state
03	Mastery estimation	Estimate knowledge, independent mastery, uncertainty and retention
04	Misconception diagnosis	Evaluate competing explanations for mistakes
05	Adaptive decision engine	Choose the next best learning action
06	Assistance and productive-struggle policy	Escalate help without creating unnecessary dependence
07	Curriculum knowledge graph integration	Respect concept dependencies and prerequisite gaps
08	Retrieval and forgetting management	Schedule and adapt future reviews
09	Teaching-strategy evaluation	Learn which interventions help under which conditions
10	Session analysis / Learning Scientist	Synthesize evidence after meaningful learning sessions
11	Assessment and answer verification contracts	Distinguish verified outcomes from unverified claims
12	Shared Learning Adaptation Model	Learn general patterns from approved, governed data
13	Privacy, safety and governance	Protect student data and constrain unsafe or unsupported decisions
14	Integration API and SDKs	Allow Qrie and other learning products to consume LAM independently
15	Evaluation and observability	Prove that LAM improves learning, not merely engagement or answer accuracy

3. Requirements for each subsystem

3.1 Evidence collection and event ledger

Purpose: build a trustworthy history of what happened during learning. LAM receives structured evidence from Qrie and other learning clients rather than relying on unstructured conversations as its primary data source.

Field	Purpose
`event_id`	Unique event identifier
`tenant_id`	Organization boundary
`student_id`	Server-validated learner identity
`session_id`	Learning-session reference
`concept_ids`	Concepts addressed by the activity
`item_id`	Question or assessment identifier
`timestamp`	Event time
`event_type`	Attempt, hint, explanation, assessment, review, etc.
`correctness`	Verified or reported outcome, with provenance
`attempt_number`	Position in the attempt sequence
`assistance_level`	Amount of assistance received
`hints_used`	Number and type of hints
`response_time_ms`	Time measurement, when available and meaningful
`self_reported_confidence`	Learner's stated confidence
`strategy_id`	Teaching strategy used
`intervention_id`	Specific intervention delivered
`difficulty`	Item difficulty or an explicitly unknown value
`assessment_mode`	Independent, assisted, delayed, transfer, or another defined mode
`schema_version`	Version of the event contract

Fields are validated against an explicit schema. Optional values stay unknown when absent; they must not be silently replaced with zero or `false`.

Requirements

- Events must be authenticated, validated and tenant-authorized.
- Retrying an event must not apply the same state update twice.
- Invalid events must be rejected or quarantined with an auditable reason.
- Event history must remain distinguishable from derived learner state.
- Corrections must be auditable.
- Concurrent events must not cause silent lost updates.
- Event ingestion must be recoverable after temporary database or service failures.
- Sensitive raw content must not be logged unnecessarily.

Acceptance test: submit the same valid event repeatedly. The ledger must preserve the required audit history while the learner-state update is applied exactly once logically.

3.2 Learner profile and state engine

Purpose: maintain a persistent, revisable representation of what a learner knows and what evidence supports that conclusion. The model is keyed by learner and concept, with additional dimensions for subject, curriculum and prerequisite relationships where relevant.

A learner's state must not be a single permanent label such as "visual learner", "weak in mathematics" or "advanced student". It evolves as new evidence arrives.

The state engine must support

- Current estimates and historical snapshots.
- Evidence-linked updates.
- Confidence or uncertainty for each relevant estimate.
- Separation between observed facts and model-derived estimates.
- State schema versioning.
- Recalculation after valid corrections.
- Concurrent update safety.
- Explanation of why a state changed.
- Unknown or insufficient-evidence states.

Recommended conceptual record (illustrative)

```
{
  "student_id": "server-resolved-id",
  "concept_id": "fractions.unlike_denominators",
  "state_version": 12,
  "mastery": {"estimate": 0.55, "uncertainty": 0.30},
  "independent_mastery": {"estimate": 0.42, "uncertainty": 0.35},
  "assisted_performance": {"estimate": 0.78, "uncertainty": 0.25},
  "evidence_count": 18,
  "last_practiced_at": "2026-10-18T14:30:00Z",
  "updated_at": "2026-10-18T14:30:00Z"
}
```

This is an illustrative schema, not a scientifically validated scoring formula. The values are placeholders; the actual model must define the meaning, calibration and permitted use of every score.

Acceptance test: given a known initial state and a fixed sequence of valid events, the engine must produce a reproducible state update, preserve its version history, and explain which evidence caused the change.

3.3 Mastery estimation and knowledge tracing

Purpose: estimate the learner's knowledge of individual concepts over time. Bayesian Knowledge Tracing (BKT) is a reasonable baseline, but it is not a complete learner model or diagnosis system. Knowledge-tracing model families differ in their assumptions, so model selection must fit the evidence and the task.

The mastery system must distinguish

- Probability or estimate of knowing a concept.
- Independent performance.
- Assisted performance.
- Evidence quality and quantity.
- Uncertainty.
- Recent performance versus persistent knowledge.
- Delayed retention.
- Ability to transfer knowledge to a new problem.

The PRD must define what counts as mastery and what evidence is sufficient to declare it. A learner might answer correctly after seeing a worked solution but fail an independent question the following day. LAM must not equate the initial assisted success with durable mastery.

Requirements

- Define a baseline mastery algorithm.
- Define how correct and incorrect responses update the state.
- Account for assistance and item difficulty.
- Define how multi-concept questions contribute evidence to each concept.
- Define when evidence is insufficient to change the estimate confidently.
- Define independent assessment and delayed assessment policies.
- Evaluate predictions and calibration on held-out data.
- Compare more sophisticated models against the baseline before deployment.

Acceptance test: a learner who repeatedly succeeds only with hints must not automatically reach the same independent-mastery status as a learner who succeeds independently on appropriate assessments.

3.4 Misconception diagnosis

Purpose: estimate why the learner is struggling and choose the most informative next step.

Diagnosis taxonomy

- Conceptual misunderstanding
- Missing prerequisite
- Procedural or arithmetic error
- Misreading the question
- Guessing
- Retrieval or memory failure
- Carelessness or isolated slip
- Incorrect reasoning strategy
- Unknown or insufficient evidence

For each candidate diagnosis, store the supporting evidence, contradicting evidence where available, confidence, status and recommended confirmation method.

Lifecycle: Candidate, then Confirmed, Rejected or Unresolved. A single wrong answer should generally be insufficient to establish a persistent misconception.

Requirements

- Maintain multiple hypotheses where appropriate.
- Use diagnostic questions when they can distinguish competing explanations.
- Do not force a diagnosis when the evidence is weak.

- Record whether a diagnosis is inferred, confirmed by an assessment, or manually reviewed.
- Revisit the diagnosis when new evidence contradicts it.
- Keep diagnosis confidence separate from mastery confidence.

Acceptance test: if the same incorrect answer could plausibly arise from a conceptual error or a calculation slip, LAM should avoid a definitive diagnosis until it receives discriminating evidence.

3.5 Adaptive decision engine

Purpose: choose the learning action most likely to improve the learner's independent capability.

Inputs the engine considers

- Current mastery and uncertainty
- Misconception candidates
- Prerequisite readiness
- Prior intervention outcomes
- Assistance dependency
- Time since practice and retention risk
- Curriculum importance and item difficulty
- Learner goals and assessment context
- Safety constraints
- Cost and latency of candidate actions

Possible actions: independent practice, a diagnostic question, a conceptual prompt, a small hint, a worked example, prerequisite remediation, a metacognitive prompt, a retrieval review, or escalation to a teacher.

The engine defines hard constraints separately from ranking preferences. For example, a prerequisite requirement may disqualify a learning activity even when that activity is otherwise highly relevant.

Every recommendation contains

Field	Meaning
`action_type`	The recommended action
`target_concept`	Concept the action addresses
`assistance_level`	Level from the assistance ladder

Field	Meaning
`strategy_id`	Teaching strategy, when applicable
`reason_codes`	Why this action won
`evidence_references`	Events and records behind the decision
`uncertainty`	How uncertain the recommendation is
`policy_version`	Decision policy that produced it
`state_version`	Learner-state version it was computed from
`fallback_action`	Alternative where needed

Acceptance test: given the same learner state, event history, graph version and policy version, the decision engine returns the same decision, unless an explicitly documented nondeterministic component is involved.

3.6 Productive-struggle and assistance policy

Purpose: provide enough support to help the learner progress without making the tutor responsible for every answer.

Assistance ladder (proposed policy structure)

Level	Name	Description
0	Independent attempt	Give the learner an opportunity to solve the problem
1	Rethink prompt	Invite another attempt or ask the learner to check their reasoning
2	Conceptual question	Target the likely gap without revealing the solution
3	Small hint	Offer one focused cue
4	Stronger hint	Provide more explicit guidance
5	Similar worked example	Demonstrate the method on a different example
6	Partial solution	Expose part of the reasoning or solution
7	Full explanation	Explain the complete method
8	Direct answer	Use only when justified by the context and policy

The actual system must allow context-appropriate transitions rather than blindly moving through every level.

Requirements

- Define when an intervention starts and when it ends.
- Define repetition limits for ineffective hints.
- Define escalation conditions.
- Define when to return to independent practice.
- Track the actual assistance delivered, not just the assistance requested.
- Distinguish help that improves immediate performance from evidence of independent learning.
- Define recovery behavior after repeated failure.
- Provide an escalation path for unresolved difficulty.

Acceptance test: repeated failure after a hint must lead to a defined alternative action, not an infinite loop of the same hint.

3.7 Curriculum knowledge graph

Purpose: ensure that LAM's recommendations respect the structure of the subject being learned. The graph defines concepts and their relationships, rather than treating all questions as isolated items.

At minimum, the contract supports

- Stable concept IDs.
- Subject, grade, curriculum and learning objective.
- Prerequisite relationships.
- Question-to-concept mapping, including multi-concept items.
- Concept and graph versions.
- Validity checks for prerequisite cycles.
- Graph lookup and dependency queries.
- Defined behavior when mappings are missing.

Qrie remains the authoritative owner of its curriculum graph unless there is a deliberate architectural reason to move that responsibility.

Acceptance test: a recommendation for an advanced concept must respect the prerequisite policy when the learner has insufficient evidence of readiness for a required prerequisite.

3.8 Retrieval scheduling and forgetting management

Purpose: bring concepts back at useful times so that learning persists beyond the current lesson. Retrieval practice has empirical support as a learning technique, including evidence of benefits for meaningful learning and inference, but LAM must still evaluate the scheduling policy for its own learners and curriculum.

Each review record includes

- Student and concept
- Last practice and last independent retrieval
- Review due time and review interval
- Review outcome and assistance level
- Item difficulty
- Delayed retention result
- Next scheduling decision
- Scheduling algorithm version

The algorithm must specify how it responds to successful recall, failed recall, assisted recall, overdue reviews, skipped reviews and repeated successful retrieval. Elapsed time alone is not proof that knowledge has been forgotten.

Acceptance test: after a failed delayed retrieval, the system must update the relevant evidence and reschedule the concept according to the documented policy.

3.9 Teaching-strategy effectiveness

Purpose: determine which strategies appear helpful for which concepts and learners. Strategies might include textual explanation, visual representation, analogy, real-world example, Socratic questioning, worked example, step-by-step explanation, diagram or story.

LAM must not permanently classify students as visual or auditory learners. It maintains evidence about strategy effectiveness in context.

Track

- Strategy and intervention IDs
- Concept and item difficulty
- Learner state before the intervention
- Assistance level
- Immediate outcome
- Subsequent independent assessment
- Delayed retention or transfer outcome

- Sample size and uncertainty
- Relevant confounding factors

Requirements

- Separate association from causal evidence.
- Avoid ranking strategies based on tiny samples.
- Use conservative defaults when evidence is insufficient.
- Allow effectiveness to vary by concept and learning objective.
- Evaluate changes against a baseline.
- Record which policy or model version selected each strategy.

Acceptance test: a strategy with two successful outcomes must not automatically be declared superior to another strategy based on a larger, differently selected sample.

3.10 Learning Scientist Agent

Purpose: synthesize meaningful learning-session evidence into a structured review of what changed and what should happen next. This component is not fully specified in the current PRD and should be added explicitly.

It produces a structured report containing

- Concepts practiced
- Demonstrated strengths and remaining weaknesses
- Evidence-backed misconception candidates
- Assisted versus independent performance
- Intervention outcomes
- Changes in confidence calibration
- Suggested learner-state updates
- Recommended future activities
- Retrieval review recommendations
- Evidence references and uncertainty

It must not become an unrestricted agent that can silently rewrite student state, modify policies or train the global model.

Requirements

- Run after meaningful sessions or defined processing triggers.
- Use structured evidence as its source of truth.
- Distinguish observations from inferences.
- Submit proposed state changes through the authoritative state engine.

- Be idempotent and recoverable.
- Support audit and reproducibility.
- Never block critical real-time learning flows unnecessarily.

Acceptance test: reprocessing the same completed session must not create duplicate learning events, duplicate reviews or conflicting state updates.

3.11 Assessment and answer verification

Purpose: establish whether learning outcomes are supported by trustworthy evidence. Qrie or another host may own the content and verifier implementation. LAM must define how verification evidence is represented and consumed.

Verifier metadata to support

- Verification source
- Assessment ID
- Item and concept IDs
- Verification method
- Verifier version
- Independent or assisted mode
- Outcome and confidence, where supported
- Evidence references
- Provisional or final status

Verification by subject

- Mathematics: symbolic or numerical checks.
- Programming: execution and tests.
- Curriculum factual questions: validation against validated curriculum material.